

Immunity

Susceptibility and Resistance

- **Susceptibility** (易感性): Host cannot eliminate parasite before it becomes established.
- **Resistance** (抗性): Host's physiological status prevents establishment and survival of parasite.
- **Immunity** (免疫): Possession of cells or tissues capable of recognizing and protecting the animal against nonself invaders.
 - **Innate immunity** (先天性免疫): Defense that does not depend on prior exposure to the invader.
 - **Acquired immunity** (获得性免疫): Specific to a particular nonself material, requires time for development, and occurs more quickly and vigorously on secondary response.

Innate Defense Mechanisms

- **Physical and Chemical Barriers** (物理和化学屏障):
 - Skin (tough, cornified, or sclerotized).
 - Mucus (lubricates, dislodges particles).
 - Antimicrobial substances: Lysozyme (溶菌酶), IgA, Interferons (干扰素), Tumor necrosis factor (TNF).
 - Low pH in stomach and vagina.
 - Commensal bacteria (inhibit pathogens).
 - **Complement** (补体): Series of proteins activated in sequence to lyse invading cells, tag them for phagocytosis (**opsonization**, 调理作用), and attract leukocytes.
- **Antimicrobial Peptides** (抗菌肽):
 - Defensins (防御素) in mammals.
 - Released upon recognition of microbial molecules by receptors like **Toll-like receptors (TLRs)**.
- **Cellular Defenses: Phagocytosis** (细胞防御：吞噬作用):
 - **Phagocytosis** (吞噬作用): Engulfing and digesting foreign particles.
 - **Phagocytes** (吞噬细胞): Monocytes (单核细胞), Macrophages (巨噬细胞), Polymorphonuclear leukocytes (PMNs, 多形核白细胞) / Granulocytes (粒细胞) (Neutrophils, Eosinophils, Basophils).
 - **Natural killer (NK) cells** (自然杀伤细胞): Kill virus-infected and tumor cells.



Figure 1: Lineage of some cells active in immune response. These cells, as well as red blood cells and other white blood cells are derived from multipotential stem cells in the bone marrow. B cells mature in bone marrow and are released into blood or lymph. Precursors of T cells complete their development in the thymus gland. Precursors of macrophages circulate in blood as monocytes.

Immunity in Invertebrates

- Recognition of nonself (allograft/xenograft rejection).
- Phagocytosis and encapsulation.
- Antimicrobial peptides.
- Some evidence of specific memory in certain invertebrates (e.g., copepods).



Figure 2: Some Invertebrate Leukocytes and Their Functions. Lists cell types and functions (Phagocytosis, Encapsulation, Allograft Rejection, Xenograft Rejection) for various invertebrate groups (Sponges, Cnidarians, Nemertines, Annelids, Sipunculids, Insects, Crustaceans, Molluscs, Echinoderms, Tunicates).

- Specific examples of invertebrate immune responses include the encapsulation and phagocytosis of parasites, as seen in resistant snails.



Figure 3: Electron micrograph of a hemocyte (H) from a schistosome-resistant strain of a snail attacking a *Schistosoma mansoni* larva (S) under in vitro conditions. Note the hemocyte processes apparently engaged in phagocytosis of portions of the larval tegument (arrows).

Acquired Immune Response in Vertebrates

- Stimulated by **Antigen** (抗原).
- Two arms:
 - **Humoral immunity** (体液免疫): Based on **antibodies** (抗体).
 - **Cellular immunity** (细胞免疫): Associated with cell surfaces.
- **Basis of Self and Nonself Recognition:**
 - **Major Histocompatibility Complex (MHC)** (主要组织相容性复合体): Proteins on cell surfaces.
 - Class I: On virtually all cells.
 - Class II: On immune cells (lymphocytes, macrophages).
- **Recognition Molecules:**
 - **Antibodies** (Immunoglobulins, Ig):
 - Structure: 2 heavy chains, 2 light chains, Y-shape. Variable region (antigen binding), Constant region.
 - Classes: IgM, IgG, IgA, IgD, IgE.
 - Functions: Opsonization, neutralization, activation of complement, Antibody-dependent cell-mediated cytotoxicity (ADCC).



Figure 4: A, Antibody molecule is composed of two shorter polypeptide chains (light chains) and two longer chains (heavy chains) held together by hydrogen and covalent disulfide bonds. These are further subdivided into variable and constant regions. B, Molecular model of antibody molecule.

- **T-Cell Receptors:** Transmembrane proteins on T cells, bind specific antigens.
- **Subsets of T Cells:**
 - **Helper T cells (TH)** (辅助 T 细胞): CD4 coreceptor. Secrete cytokines. TH1 (cell-mediated), TH2 (humoral).
 - **Cytotoxic T lymphocytes (CTLs)** (细胞毒性 T 细胞): CD8 coreceptor. Kill target cells.
 - **Suppressor T cells** (抑制 T 细胞).
 - **Memory T cells** (记忆 T 细胞).
- **Cytokines** (细胞因子): Protein hormones for communication (e.g., Interleukins, Interferons, TNF).

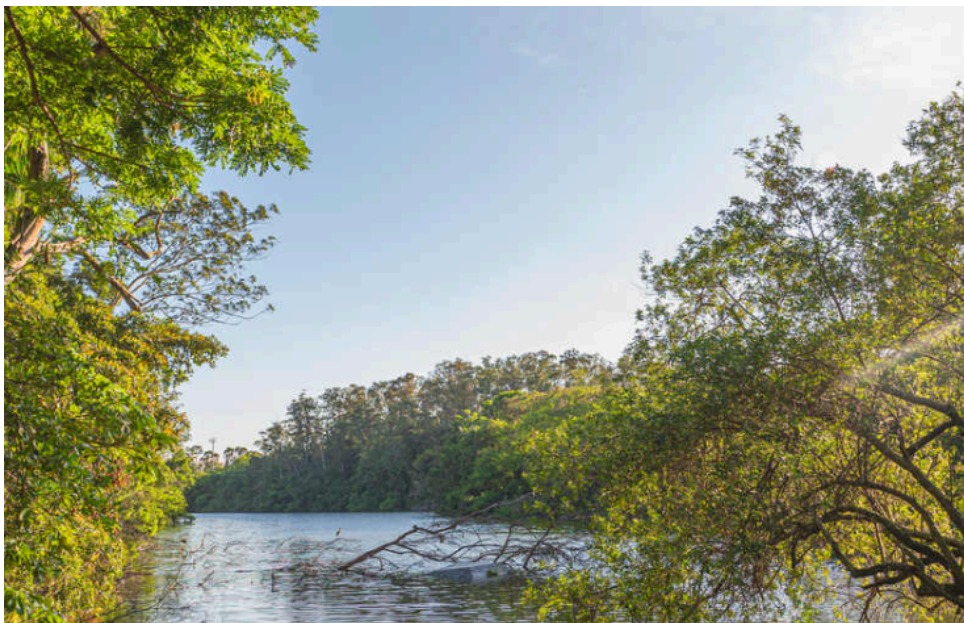


Figure 5: Major pathways involved in cell-mediated (TH1) and humoral (TH2) immune responses as mediated by cytokines. Solid arrows indicate positive signals and broken arrows indicate inhibitory signals. Broken lines without arrows indicate path of cellular activation.

- Cytokines are diverse and have specific sources and functions, regulating the immune response.



Figure 6: Some Important Cytokines. Lists Principal Source and Function for Interleukin-1 (IL-1), IL-2, IL-3, IL-4, IL-5, IL-6, IL-8, IL-10, IL-12, Transforming growth factor-beta (TGF-beta), Interferon-alpha (IFN-alpha), Interferon-beta (IFN-beta), Interferon-gamma (IFN-gamma), and Tumor necrosis factor (TNF).

Generation of Humoral Response (TH2 Arm)

- Antigen binds to B cell surface antibody.
- B cell internalizes and presents **epitope** (抗原决定簇) with MHC II.
- TH2 cell recognizes epitope-MHC II complex (aided by CD4).
- TH2 secretes cytokines (IL-4, IL-5, etc.) to activate B cell.
- B cell proliferates into **Plasma cells** (浆细胞) (secrete antibody) and **Memory cells** (记忆细胞).



Figure 7: Humoral immune response. (1) Surface antibodies of the B cell bind antigen. (2) TH2 cell recognizes antigen and class II protein on the B cell. (3) TH2 stimulates proliferation of the B cell. (4) Cytokines promote activation and differentiation. (5) Plasma cells secrete antibodies. (6) Memory cells are formed. (7) Antibody binds to antigen and stimulates macrophages (opsonization).

- The interaction between the B cell and the Helper T cell involves specific surface molecules.

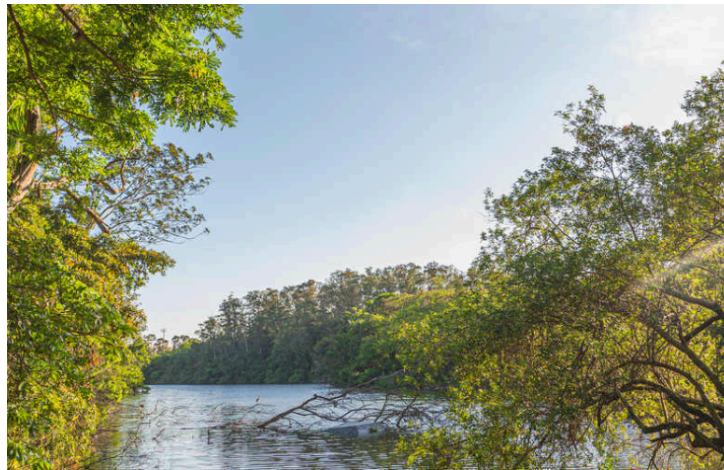


Figure 8: Interacting molecules during activation of a TH2-helper cell by an antigen-presenting B cell. Shows Peptide epitope, Costimulatory molecules, CD4, MHC class II, T-cell receptor, CD40, and CD40 receptor.

- **Secondary response** (二次免疫应答): Faster and stronger due to memory cells.

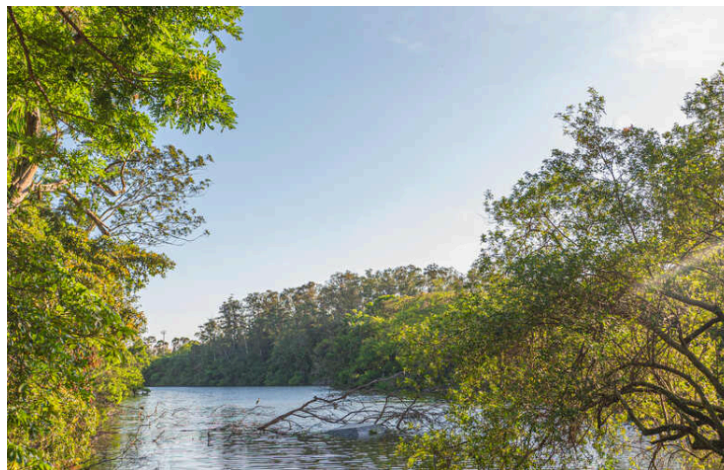


Figure 9: Typical antibody response after primary and challenge immunizations. The secondary response is a result of the large numbers of memory cells produced after the primary B-cell activation.

Cell-Mediated Response (TH1 Arm)

- Involves TH1 cells, CTLs, macrophages.
- TH1 cells recognize epitope-MHC II on APCs.
- Secrete IL-2, TNF, IFN-gamma.
- Activate CTLs (kill infected cells), NK cells, and macrophages.
- Important for intracellular parasites (viruses, bacteria) and tumor cells.
- Rejection of tissue transplants (suppressed by drugs like cyclosporine).

Inflammation

- **Inflammation** (炎症): Nonspecific response to tissue damage or infection.
- Signs: Redness, swelling, heat, pain.
- Mediators: Histamine (from mast cells), TNF, cytokines.
- Vasodilation and increased permeability allow leukocytes to enter tissue.

- **Delayed type hypersensitivity (DTH):** Cell-mediated (macrophages).

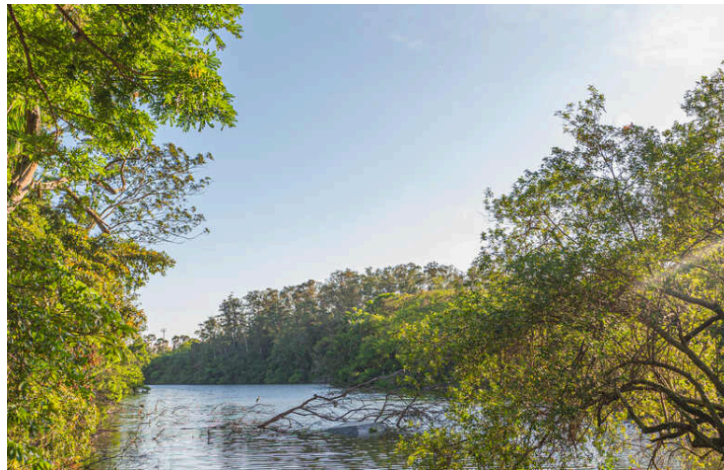


Figure 10: Granulomatous reaction around eggs (arrows) of *Schistosoma mansoni* in mesenteries.

- **Immediate hypersensitivity:** Antibody-mediated (IgE, mast cells). Basis of **Allergy** (过敏) and **Asthma** (哮喘).

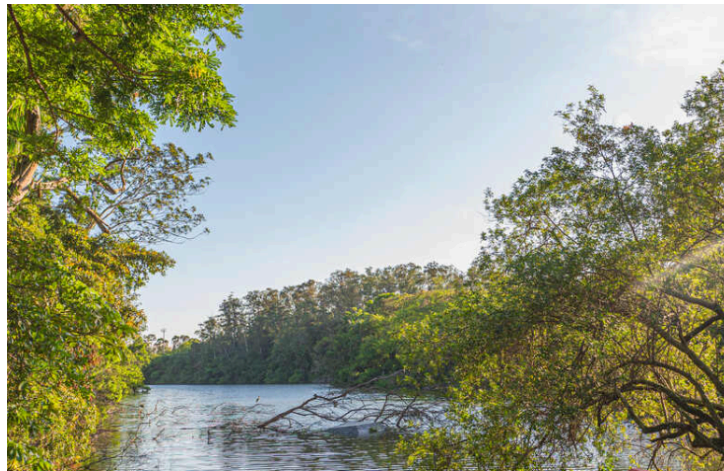


Figure 11: Wheal and flare reactions around sites of antigen injection for allergy testing.

Acquired Immune Deficiency Syndrome (AIDS)

- Caused by **HIV** (Human Immunodeficiency Virus).
- Destroys TH cells (CD4 receptor).
- Disables humoral and cell-mediated immunity.

Blood Group Antigens

- **ABO Blood Types:** Antigens A and B on erythrocytes.
 - Type A (A antigen, anti-B antibody).
 - Type B (B antigen, anti-A antibody).
 - Type AB (A and B antigens, no antibodies) - Universal recipient.
 - Type O (No antigens, anti-A and anti-B antibodies) - Universal donor.
- **Rh Factor:** Rh positive (has antigen) vs. Rh negative.
 - **Erythroblastosis fetalis** (新生儿溶血症): Rh- mother sensitized by Rh+ fetus, antibodies attack fetus in subsequent pregnancy.



Figure 12: Major Blood Groups. Lists Blood Type, Genotype, Antigens on Red Blood Cells, Antibodies in Serum, Can Give Blood To, Can Receive Blood From, and Frequency in United States (Whites, Blacks, Asians).